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Dynamic Target Assignment and Cooperative Decision-Making for UAV Swarms Based on Multi-Agent Reinforcement Learning

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***Abstract*—Safe and efficient cooperative flight in cluttered, dynamic multi-target environments requires UAV swarms to repeatedly reassign targets while maintaining collision-free trajectories under partial observability. Existing methods often decouple target assignment and path planning into hand-engineered pipelines, which are effective in static, fully known settings but become brittle when targets move and perception is uncertain, leading to reduced cooperation efficiency and poor adaptability. This paper proposes DA-MAPPO, a dynamic assignment-aware multi-agent reinforcement learning framework built on PPO. DA-MAPPO integrates an online minimum-cost target allocation module into a unified perception–allocation–decision-making loop: real-time assignment outcomes are embedded into each agent's local observation to form assignment-augmented states, enabling step-by-step coordination reconfiguration. In addition, a hierarchical cooperative reward is designed to jointly encourage individual safety (collision avoidance) and team-level mission efficiency. High-fidelity simulation results in Gazebo show that DA-MAPPO achieves 90%–99% mission success across dynamic multi-target scenarios with varying obstacle densities, outperforming representative baselines by up to 25 percentage points. When transferring from static to dynamic targets, DA-MAPPO exhibits negligible degradation in low- and medium-density environments and only about a 2% drop in the densest setting, while baselines degrade substantially. DA-MAPPO also produces shorter average trajectories and fewer decision steps, indicating higher mission efficiency with lower collision risk.**

***Index Terms*—UAV Swarms, Multi-Agent Reinforcement Learning, Dynamic Target Allocation, Cooperative Navigation**

I. Introduction

UNMANNED aerial vehicle (UAV) swarms are emerging as a core technological paradigm for wide-area monitoring, rapid response, and distributed operation at the edge of the Internet of Things (IoT) [1], [2]. By leveraging distributed perception, cooperative decision-making, and emergent swarm intelligence [3], [4], multi-UAV systems deliver robustness, flexibility, and efficiency that single-agent platforms cannot achieve in applications such as smart-city management [5], large-scale environmental monitoring [6], and emergency response [7]. In many complex swarm tasks, such as cooperative dynamic target tracking, area security containment, and intelligent adversarial gaming [8], [9], the environment is highly dynamic, and the states of multiple targets change over time [10]. Effective operation relies on two tightly coupled capabilities: (i) efficient multi-target assignment and (ii) safe, collision-free navigation under partial observability and bandwidth-limited inter-agent communication [11], [12]. What is especially important is that, without a central global node for coordination, the swarm must rely on local sensing and shared communication between individuals to spread key information, enabling it to accomplish complex tasks through a distributed decision-making process [13].

Traditional approaches, grounded in optimization theory, typically decouple the problem into target assignment and path planning, and solve the two sequentially with a centralized controller and a global state [14], [15]. While these methods can achieve theoretical optimality in deterministic environments, their reliance on global information and high computational complexity hinder adaptability to dynamically changing conditions. In recent years, Multi-Agent Deep Reinforcement Learning (MADRL) has emerged as a promising alternative for distributed decision-making. Through end-to-end learning, MADRL enables direct learning of cooperative policies from perceptual inputs, demonstrating remarkable adaptability in dynamic environments [16]–[18]. However, existing MADRL-based studies exhibit notable limitations: most presume static target positions, thus failing to address practical scenarios with mobile targets, while often oversimplifying swarm coordination tasks without achieving effective real-time perception-driven decision-making. Beyond cooperative navigation and task allocation, recent studies on large-scale UAV networking and distributed aerial-vehicular systems further highlight that practical swarm intelligence must account for communication scalability, network robustness, and edge-assisted coordination [19], [20]. In addition, UAV-assisted federated learning over massive MIMO has highlighted the broader role of UAVs in edge intelligence and communication-efficient learning, which is complementary to our focus on online target reassignment and safe cooperative navigation [21].

Against this background, we propose DA-MAPPO, a dynamic assignment-aware multi-agent proximal policy optimization framework that couples online target allocation with decentralized control learning. At each decision step, an online allocator updates one-to-one agent–target pairings using a minimum path-cost principle, and the allocation outcome is fused with each agent's local sensing/communication observations to construct assignment-augmented states. A hierarchical cooperative reward further balances individual safety and efficiency with team-level coordination, enabling adaptive reassignment and collision-averse navigation under local observations and lightweight inter-agent communication.

We evaluate DA-MAPPO in a Gazebo-based high-fidelity simulator across static and dynamic multi-target scenarios with

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varying obstacle densities. DA-MAPPO achieves 90%–99% success rates in dynamic environments and improves success rate by up to 25 percentage points over strong baselines. Moreover, when targets switch from static to dynamic, DA-MAPPO shows almost no degradation in low/medium densities and only a 2% drop in the densest setting, demonstrating robust adaptability for edge-IoT swarm deployments.

In summary, the main contributions of this paper are as follows:

1) We develop a unified framework that integrates an online minimum-cost target allocator with MAPPO and constructs assignment-augmented observations, enabling coupled optimization of task allocation and obstacle-aware decentralized navigation in dynamic multi-target environments.
2) We design a multi-term hierarchical reward that jointly accounts for individual safety and efficiency as well as team-level coordination, promoting collision-averse, compact, and time-efficient swarm behaviors while supporting stable policy learning.
3) We build a Gazebo-based high-fidelity testbed and conduct comprehensive comparisons with representative baselines in both static and dynamic multi-target scenarios.

II. Related Work

A. Optimization-Based Task Assignment and Path Planning

Multi-UAV mission planning has long relied on optimization-based pipelines that separately or jointly solve task/target assignment and trajectory planning. In IoT-assisted UAV networks, assignment objectives may include coverage quality, load balancing, or timeliness metrics such as Age of Information (AoI), and are often coupled with trajectory optimization to improve communication/sensing performance [22], [23]. For disaster rescue and time-critical missions, joint mission assignment and 3D path planning have also been studied to satisfy feasibility and mission efficiency constraints [24]. In adversarial settings (e.g., malicious jamming), researchers further consider joint optimization across power, path, and target association to enhance robustness [25].

A large body of work adopts metaheuristics to handle combinatorial complexity and heterogeneous constraints. Typical examples include genetic algorithms for cooperative allocation with simultaneous-arrival and resource constraints [26], hybrid schemes combining sampling-based planning and swarm intelligence (e.g., GDRRT*-PSO) [27], and bio-inspired strategies such as wolf-pack methods [28]. Ant-colony variants and other metaheuristics are widely used for coverage and formation/heterogeneous mission planning [29]–[31], and improved heuristic optimization has been applied to multi-task path planning [32]. Although these methods can achieve strong performance in static or quasi-static settings, they usually depend on centralized/global information and repeated replanning; when targets move frequently or assignments must be updated online, the decoupling between assignment and real-time collision-averse control can lead to brittleness and latency issues especially under partial observability and limited onboard communication.

B. Task-Driven DRL for Multi-UAV Coordination and Allocation

Deep reinforcement learning has increasingly been used to address multi-UAV coordination in complex environments due to its capability of learning reactive policies. Recent studies have explored PPO-based improvements for collaborative decision-making [33], hierarchical DRL for large-scale scheduling and confrontation tasks under uncertainty [34], [35], and scalable MARL through improved information aggregation [36]. Digital-twin-assisted DRL has also been introduced to enhance training efficiency and realism for multi-UAV task assignment [37]. Beyond pure learning, hybrid formulations combine RL with classical search/optimization to improve task allocation and planning performance [38]. In the IoT context, DRL has been applied to data collection and mission execution with multi-UAV coordination [39].

For navigation and multi-target tasks, RL has been used for autonomous tracking and two-stage decision frameworks that incorporate expert experience [40], and for multi-UAV path planning/following via MARL [41]. Some works further embed RL guidance into metaheuristics (e.g., grey wolf optimization guided by RL) to balance exploration and solution quality [42]. Meanwhile, task decomposition and algorithmic variants such as TD3-style approaches have been adopted for autonomous path planning [43].

Despite this progress, many task-driven MARL approaches implicitly assume that target assignments are fixed or only weakly coupled to the learned control policy. Even when allocation is considered, it is often handled as a separate module or a pre-processing step, making it harder to support frequent online reassignment in dynamic multi-target scenes while guaranteeing safe multi-agent navigation.

C. Perception-Driven MADRL for Autonomous Navigation

Another line of work focuses on decentralized navigation and collision avoidance based on onboard perception and limited communication. Multi-agent DRL has been used for navigation in unknown complex environments [44], and perception-driven collision avoidance has been studied using vision-based distributed DRL [45]. To improve coordination, learning-based flocking and repulsion mechanisms have been proposed under limited visual fields [46]. In network-constrained environments, researchers consider cooperative navigation under communication coverage constraints [47], as well as partially observable coverage maintenance via graph reinforcement learning [48]. More recently, network-constrained flocking and obstacle avoidance policies have been developed for multi-UAV systems [49], and efficient RL-based cooperative navigation in complex environments has also been explored [50]. For target-coverage tasks in dynamic environments, learning/planning methods have been investigated to balance coverage and feasibility [51]. These perception/network-constrained methods strengthen local collision avoidance and decentralized execution, but they often do not treat dynamic target assignment as a first-class decision that must be updated online. Consequently, when multiple moving targets require persistent reassignment, there remains a gap in integrating assignment dynamics with decentralized continuous control under partial observability and communication limits.

D. Summary

In summary, optimization-based approaches provide principled assignment and planning but tend to rely on centralized information and repeated replanning, which can be brittle under dynamic targets and partial observability. DRL/MADRL improves reactivity and decentralization, and has been applied to allocation, scheduling, navigation, and coverage under IoT/network constraints, yet online reassignment coupled with safe decentralized control is still insufficiently addressed, especially in realistic dynamic multi-target settings. This motivates our DA-MAPPO framework that integrates online assignment with MARL through assignment-augmented observations and cooperative reward shaping.

III. PROBLEM FORMULATION

A. Unmanned Swarm Task Scenario Description

Within IoT-edge environments, UAV swarms operate as distributed, perception-driven agents capable of collaboratively executing complex tasks such as target pursuit, interception, and data collection. These missions frequently involve multiple mobile targets whose states evolve continuously over time, making static pre-assignment strategies fundamentally insufficient. Such fixed allocation schemes are unable to adapt to time-varying target movements, leading to inefficient trajectories, increased collision risks, and degraded overall mission performance.

As shown in Fig. 1, consider a swarm of N unmanned aerial vehicles (UAVs), denoted by the set $\mathcal{N} = \{1, \dots, n\}$, operating within a bounded 3D workspace $W \subset \mathbb{R}^3$. At any decision time step t, the state of agent $i \in \mathcal{N}$ is defined by its position vector $\mathbf{s}_i(t) = [x_i(t), y_i(t), z_i(t)]^T \in W$. To maintain computational efficiency for edge-IoT deployments and focus on cooperative maneuvers, we impose a strict altitude constraint:

$$z_i(t) = H, \quad \forall i \in \mathcal{N}, \forall t \geq 0, \tag{1}$$

where H is a constant flight altitude. This constraint effectively reduces the operational configuration space to a 2D manifold $\mathcal{W} \subset \mathbb{R}^2$. To formally define the mission environment, the workspace $\mathcal{W}$ is partitioned into three disjoint functional sets:

- Takeoff area ($\mathcal{W}_{init} \subset \mathcal{W}$): The region where UAVs initialize and take off, ensuring independence from subsequent navigation processes.
- Obstacle area ($\mathcal{O}_{obs} \subset \mathcal{W}$): The main test area is populated with static obstacles whose density controls task difficulty.
- Dynamic Target Set (G): All target points are randomly generated within this region.

The overall mission objective is to achieve real-time adaptive target allocation and collision-free navigation based solely on local perception and limited inter-UAV communication. Formally, the swarm navigation environment is modeled as

$$E = \left(W, \mathbf{s}^u, G, O^{\text{obstacle}}, C\right) \tag{2}$$

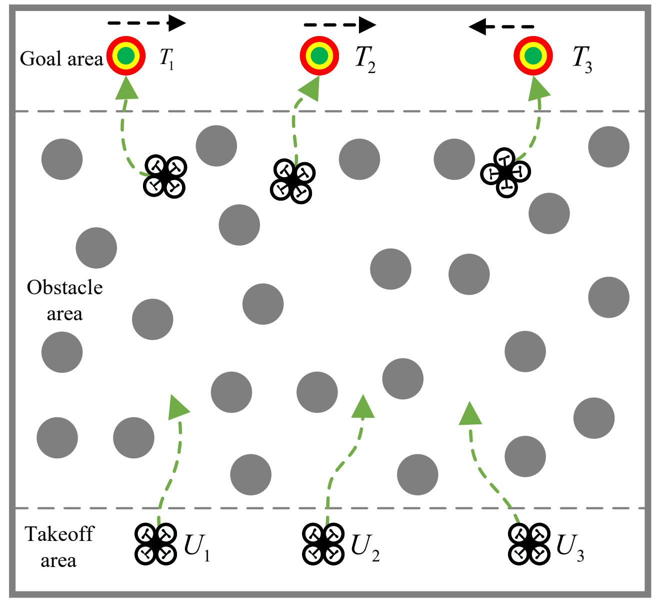

Fig. 1. UAV Swarm Mission Scenario.

The target region for the swarm mission is formulated as

$$G = \left\{\mathbf{s}^u \in W^N \;\middle|\; \left\|\mathbf{s}^u - \mathbf{s}^g\right\| \leq \varepsilon_0\right\} \tag{3}$$

where $\mathbf{s}^g$ denotes the target position and $\varepsilon_0 > 0$ represents the arrival threshold.

The mission objective can be formulated as finding an optimal strategy π^* minimizing total navigation time T while satisfying safety, collision, and communication constraints:

$$\begin{gathered}
\min_{\pi} \; T = \sum_{i=1}^{N} \sum_{t=0}^{T_f} \mathbb{I}(\|s_i^u(t) - s_i^g(t)\| \leq \epsilon_o) \\
\text{s.t.} \\
\left\|s_i^u(t) - s_i^o(t)\right\| \geq d_{\text{safe}} \\
\left\|s_i^u(t) - s_j^u(t)\right\| \geq d_{\text{col}}, \;\; i \neq j \\
\left\|s_i^u(t) - s_j^u(t)\right\| \leq R_{\text{com}}, \;\; i \neq j
\end{gathered} \tag{4}$$

where $\mathbf{s}_i^o$ denotes the obstacle position, while d_{safe} and d_{col} specify the safety buffers for obstacles and inter-UAV separation, respectively. To account for physical dimensions and kinematic constraints, each UAV is modeled as a circular rigid body . The safety threshold is analytically defined as $d_{safe} \geq v_{max}\Delta t + \frac{v_{max}^2}{2a_{max}}$ which represents the minimum distance required for a UAV to come to a complete stop within one control cycle. This conservative modeling approach ensures that any collision-free trajectory under the abstraction provides a reliable upper bound for safety in the underlying physical system.

B. Local and Imperfect Inter-UAV Communication Model

In practical UAV swarm deployments, inter-agent communication is typically constrained by limited bandwidth, communication range, and network unreliability. To faithfully capture these characteristics, we model the swarm communication as a local, distance-limited, and potentially imperfect information-sharing mechanism.

Specifically, at each decision time step t, UAV i can communicate only with neighboring UAVs located within a finite communication radius R_{com}. The resulting communication topology is modeled as a dynamic undirected graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$, where each vertex represents a UAV and an edge $(i,j) \in \mathcal{E}_t$ exists if and only if $\left\|\mathbf{p}_i^t - \mathbf{p}_j^t\right\| \le R_{\text{com}}$.

The exchanged communication content is limited to lightweight state information, specifically the relative positional information between neighboring UAVs. No global state, raw sensor observations, or absolute target positions are shared. The received communication messages are aggregated into a local swarm topology feature $\mathbf{q}_i^t$, which encodes the relative spatial configuration of neighboring agents and is incorporated into the local observation of UAV i. For UAV i, if k_i neighboring messages are received at time step t, the local information fusion complexity is $O(k_i)$, with worst-case $O(N-1)$ under the fixed-length topology representation adopted in this work. Because the exchanged data consist only of low-dimensional relative positions, the fusion step remains lightweight and does not involve graph optimization, iterative consensus, or global state reconstruction, thereby supporting decentralized execution on resource-constrained edge-IoT platforms.

Furthermore, the communication channel is not assumed to be ideal. To reflect real-world network conditions, stochastic packet loss and communication latency are introduced during evaluation, as detailed in the robustness analysis. Importantly, the proposed framework does not rely on instantaneous or perfect communication, ensuring that decentralized execution remains feasible under degraded network conditions.

C. MADRL-Based Autonomous Decision Modeling

To address the decentralized nature of swarm decision-making, the cooperative navigation problem is formulated as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP): $\langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{O}, P, \mathcal{R}, \gamma \rangle$, where $\mathcal{N} = \{1, 2, \ldots, n\}$ is the agent set, $\mathcal{S}$ and $\mathcal{A} = \{A_1, A_2, \ldots, A_n\}$ are the global state and joint action spaces, $\mathcal{O} = \{O_1, O_2, \ldots, O_n\}$ is the joint observation space, P represents the state transition probability density function, $\mathcal{R}$ is the global reward, and $\gamma \in [0, 1)$ is the discount factor.

Each agent executes a local policy: $\pi_i : O_i^t \to A_i$ mapping its local observation o_i^t to the action output $a_t^i \sim \pi_i(\cdot \mid o_t^i)$. The objective of multi-agent reinforcement learning is to find an optimal joint policy π^* that maximizes the expected cumulative discounted return:

$$\pi^* = \arg\max_{\pi} J(\pi)$$

$$J(\pi) = \mathbb{E}_{s_0 \sim \rho_0,\, a_t \sim \pi(\cdot|o_t),\, s_{t+1} \sim P(\cdot|s_t, a_t)} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right] \tag{5}$$

where ρ_0 represents the initial state distribution.

The state-value function and action-value function are formally defined as:

$$V^\pi(s_t) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{R}(s_{t+k}, a_{t+k}) \right] \tag{6}$$

$$Q^\pi(s_t, a_t) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{R}(s_{t+k}, a_{t+k}) \right] \tag{7}$$

The advantage function is employed to evaluate the relative value of an action:

$$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t) \tag{8}$$

This theoretical formulation establishes the foundation for the multi-agent reinforcement learning strategy utilized in the proposed DA-MAPPO framework.

IV. METHODOLOGY

A. Framework Overview

As discussed in Section II, a fundamental limitation of existing approaches is the decoupling of target assignment from real-time navigation control, which results in brittle coordination under dynamic conditions. To bridge this gap, we propose DA-MAPPO, an adaptive cooperative navigation framework that tightly integrates dynamic target allocation into the perception–decision loop. The central design principle is that embedding up-to-date assignment outcomes directly into each agent's observation enables joint optimization over the allocation–navigation objective through end-to-end policy learning, rather than treating them as independent subproblems. As illustrated in Fig. 2, the resulting architecture forms a perception–allocation–decision closed loop that supports autonomous cooperative navigation and obstacle avoidance under continuously evolving target configurations.

At each decision time step t, the framework executes three stages as follows:

1) Environmental Perception: Each UAV acquires local environmental observations $\mathbf{z}_t^i$ through onboard LiDAR, exchanges positional information $\mathbf{q}_t^i$ with neighboring UAVs via a constrained communication network, and combines this with real-time kinematic data $\mathbf{u}_t^i$ to construct an initial observation vector $\mathbf{o}_t^{\text{raw},i} = [\mathbf{z}_t^i,\, \mathbf{u}_t^i,\, \mathbf{q}_t^i]$
2) Target Allocation: Based on the current positions of UAVs and targets, an adaptive allocation algorithm computes the optimal task assignment strategy $\mathcal{A}_t^* = \Phi\big(\mathbf{s}_t^u,\, \mathbf{s}_t^g\big)$ dynamically assigning dedicated pursuit targets to each UAV. This allocation strategy aims to minimize global path cost while satisfying one-to-one task assignment constraints.
3) Cooperative Decision-Making: Each UAV uses its assigned target information with the initial observations to obtain an enhanced local observation: $\mathbf{o}_t^i = [\mathbf{z}_t^i,\, \mathbf{u}_t^i,\, \mathbf{g}_t^i,\, \mathbf{q}_t^i]$. A MADRL-based policy network then generates continuous control commands $\mathbf{a}_t^i$. These commands drive the UAV's dynamic system to navigate toward designated targets while avoiding obstacles.

This closed-loop process executes at every decision step throughout the mission. As targets reposition, the allocation is updated accordingly, enabling the swarm to continuously adapt its cooperative strategy. The tight coupling between allocation and policy learning distinguishes DA-MAPPO from prior methods that treat assignment and control as loosely connected modules.

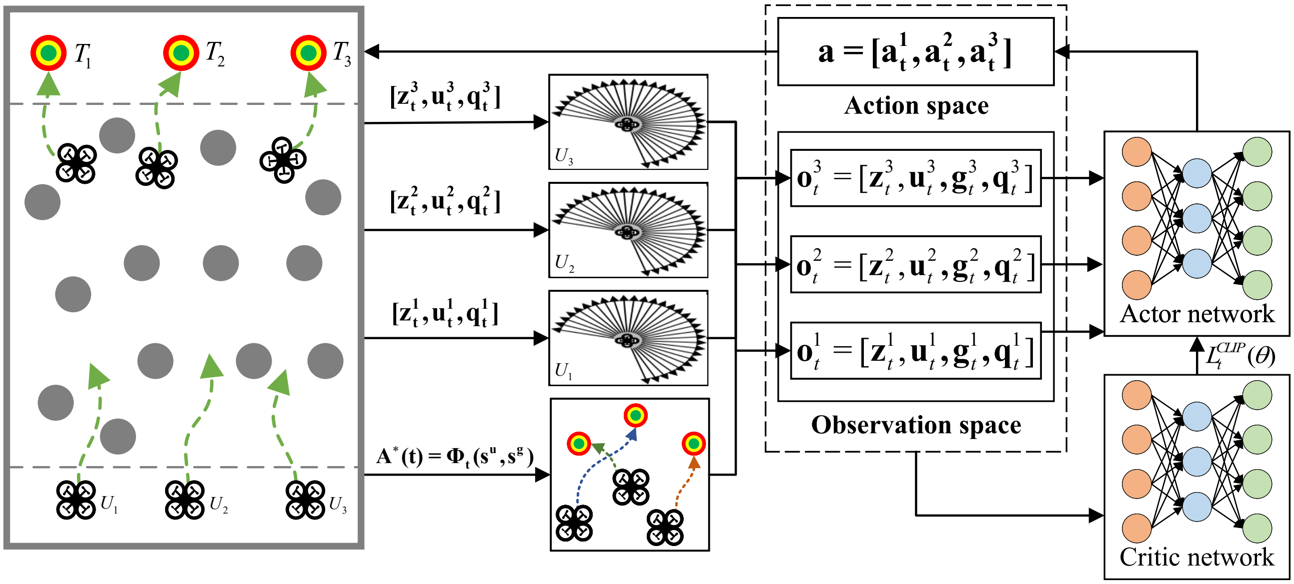

Fig. 2. An Adaptive Cooperative Navigation Framework for UAV Swarms in Dynamic Multi-Target Scenarios.

B. Agent Observation and Action Representation

Under the Dec-POMDP formulation introduced in Section III-C, each agent acts on locally available information. A structured observation vector is therefore designed to encode the heterogeneous perceptual and cooperative cues required for decentralized decision-making. Specifically, the observation of the i-th agent is defined as:

$$\mathbf{o}_t^i = [\mathbf{z}_t^i,\ \mathbf{u}_t^i,\ \mathbf{g}_t^i,\ \mathbf{q}_t^i] \in \mathbb{R}^{d_{\text{loc}}} \tag{9}$$

The four sub-vectors capture complementary aspects of each agent's situational awareness:

Obstacle perception $\mathbf{z}_t^i \in \mathbb{R}^D$: A 2D LiDAR provides $D{=}35$ range measurements at a fixed angular resolution, encoding the local obstacle geometry.

Ego-motion state $\mathbf{u}_t^i = [v_t^i, \omega_t^i, a_{v,t}^i, a_{\omega,t}^i] \in \mathbb{R}^4$: Forward linear velocity, yaw angular velocity, and their respective accelerations. The accelerations are obtained from finite-difference estimates smoothed by a Kalman filter to suppress sensor noise.

Target features $\mathbf{g}_t^i = [d_t^i, \Delta\theta_t^i] \in \mathbb{R}^2$: As shown in Fig. 3, the assigned target is encoded in a body-frame polar representation, where $d_t^i = \|\mathbf{p}_t^{u_i} - \mathbf{s}_t^{g_i}\|_2$ is the Euclidean distance and $\Delta\theta_t^i = \operatorname{arctan2}(y_t^{g_i} - y_t^{u_i},\, x_t^{g_i} - x_t^{u_i}) - \Psi_t^i$ is the relative bearing with respect to the UAV heading Ψ_t^i. This compact representation provides the policy with both direction and proximity information about the current goal.

Swarm topology $\mathbf{q}_t^i \in \mathbb{R}^{2(N-1)}$: The relative positions $(\Delta x_t^{i,j}, \Delta y_t^{i,j})$ of all teammates, concatenated into a fixed-length vector that informs the policy about the spatial configuration of the swarm.

Combining all components yields $d_{\text{loc}} = D + 4 + 2 + 2(N{-}1)$. Each agent outputs a two-dimensional continuous control command $\mathbf{a}_t^i = (v^i, \omega^i)$ with $v^i \in [-1, 1]$ m/s and $\omega^i \in [-1, 1]$ rad/s, providing sufficient control authority for agile maneuvering while regularizing the policy output to facilitate stable learning.

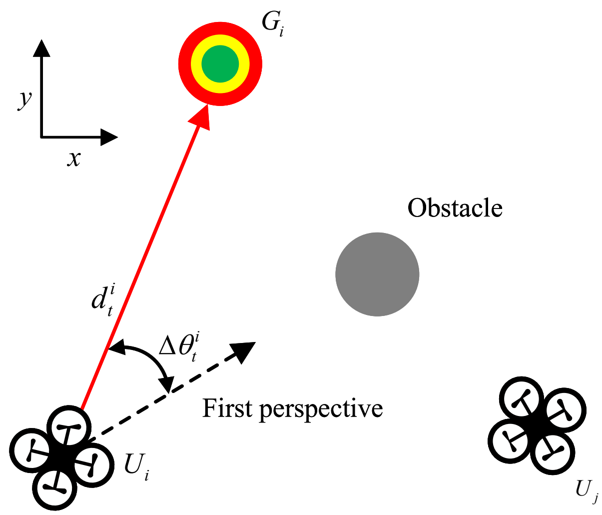

Fig. 3. Spatial Relationship Between UAV and Target.

An episode terminates under one of four conditions: (i) target arrival, triggered when $\|\mathbf{p}_t^{u_i} - \mathbf{s}_t^{g_i}\|_2 \le \varepsilon_0$ for all agents; (ii) workspace boundary violation; (iii) collision, detected either through LiDAR range readings falling below d_{safe} (obstacle collision) or through inter-agent distance falling below d_{col}; or (iv) exceeding a maximum step limit $T_{\max}$. If any single agent triggers a safety violation (boundary, collision, or timeout), the entire episode terminates immediately to preserve task integrity. The episode is deemed successful only when all agents satisfy the arrival criterion simultaneously.

C. Reward Function Design

Reward shaping is critical in MARL, as the reward signal must simultaneously guide individual behavior and promote emergent team-level coordination. A poorly structured reward can lead to conflicting gradients among agents, credit-assignment ambiguity, or convergence to locally optimal but globally inefficient joint policies. We therefore adopt a hierarchical decomposition that separates the reward into four functionally distinct tiers—team coordination, individual navigation, safety constraints, and auxiliary regularization—each targeting a specific aspect of the desired swarm behavior.

1) Team Coordination Reward: To incentivize globally coordinated progress rather than individually greedy behavior, a shared potential-based reward penalizes the aggregate distance between the swarm and its assigned targets:

$$R_{\mathrm{team},t} = -\omega_{\mathrm{team}} \sum_{j=1}^{N} d_t^j \tag{10}$$

where d_t^j denotes the Euclidean distance between UAV j and its corresponding target point, and $\omega_{\mathrm{team}} > 0$ is the team reward weight coefficient.

2) Individual Navigation Reward: While the team reward drives collective behavior, each agent additionally receives individual signals that shape its local navigation strategy. The progress reward provides a dense, step-wise gradient based on the change in target distance:

$$R_{\mathrm{prog},t}^i = -\kappa_{\mathrm{prog}} \left(d_{t-1}^i - d_t^i \right) \tag{11}$$

where $\kappa_{\mathrm{prog}} > 0$ is the progress coefficient.

To address the sparse-reward challenge inherent in goal-reaching tasks, a one-time arrival bonus is granted upon first entry into the target region:

$$R_{\mathrm{arr},t}^i = \begin{cases} b_k, & d_t^i < \varepsilon_0 \wedge \neg \mathrm{arrived}_i, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

where b_k is a monotonically decreasing bonus sequence indexed by the arrival order k, encouraging early arrivals and mitigating the free-rider effect, and $\mathrm{arrived}_i$ tracks whether UAV i has already collected this bonus.

Once an agent reaches its target, a stationarity maintenance reward discourages post-arrival drift and ensures stable hovering:

$$R_{\mathrm{hover},t}^i = \begin{cases} +1.0, & d_t^i < \varepsilon_0, \\ -10.0 \left(d_t^i - \varepsilon_0 \right), & d_t^i \geq \varepsilon_0 \wedge \mathrm{arrived}_i, \\ 0, & \text{otherwise}. \end{cases} \tag{13}$$

3) Safety Constraint Reward: Collision avoidance is enforced through a graded penalty mechanism that increases in severity as an agent approaches danger. For obstacle proximity, a piecewise hard penalty is defined over concentric safety zones around each obstacle:

$$R_{\text{laser-hard},t}^i = \begin{cases} -100.0, & \min(z_t^i) < d_{\mathrm{safe}}, \\ -25.0, & d_{\mathrm{safe}} \leq \min(z_t^i) < 2d_{\mathrm{safe}}, \\ -10.0, & 2d_{\mathrm{safe}} \leq \min(z_t^i) < 3d_{\mathrm{safe}}, \\ 0, & \min(z_t^i) \geq 3d_{\mathrm{safe}}. \end{cases} \tag{14}$$

To provide a continuous gradient signal in the proximity zone, an additional soft penalty accumulates a linear cost over all LiDAR beams closer than $2d_{\mathrm{safe}}$:

$$R_{\text{laser-soft},t}^i = \sum_{r \in z_t^i,\, r < 2d_{\mathrm{safe}}} 0.5 \left(r - 2d_{\mathrm{safe}} \right) \tag{15}$$

Inter-agent collision is penalized symmetrically with a large negative reward upon violation of the minimum separation constraint:

$$R_{\mathrm{inter},t}^i = -100.0\, \mathbb{I}\left(\min_{i \neq j} \left\| \mathbf{p}_t^{u_i} - \mathbf{p}_t^{u_j} \right\|_2 \leq d_{\mathrm{col}} \right) \tag{16}$$

The aggregate safety penalty for agent i is the sum of the above three components:

$$R_{\mathrm{safe},t}^i = R_{\text{laser-hard},t}^i + R_{\text{laser-soft},t}^i + R_{\mathrm{inter},t}^i \tag{17}$$

4) Auxiliary Regularization Terms: Three auxiliary terms regularize the learned policy to promote smooth, efficient, and bounded behavior. An action smoothness penalty discourages abrupt control changes by penalizing body-frame accelerations:

$$R_{\mathrm{acc},t}^i = -\alpha_v \left| \dot{v}^i \right| - \alpha_\psi \left| \dot{\omega}^i \right| \tag{18}$$

where $\alpha_v = 0.5, \alpha_\psi = 0.2$.

A constant per-step penalty encourages time-efficient task completion:

$$R_{\mathrm{step},t}^i = -1.0 \tag{19}$$

Finally, a boundary penalty confines agents within the workspace:

$$R_{\mathrm{bdry},t}^i = \begin{cases} -100.0, & d_{\mathrm{bdry}} < 0, \\ -\gamma_b \left(1 - \dfrac{d_{\mathrm{bdry}}}{\omega_b} \right), & 0 \leq d_{\mathrm{bdry}} < \omega_b, \\ 0, & \text{otherwise}, \end{cases} \tag{20}$$

where $\gamma_b = 100.0$.

The per-agent reward at each step aggregates all four tiers with a global normalization factor:

$$\begin{aligned} R_t^i = \frac{1}{50} \Big(& R_{\mathrm{team},t} + R_{\mathrm{prog},t}^i + R_{\mathrm{arr},t}^i + R_{\mathrm{hover},t}^i \\ & + R_{\mathrm{safe},t}^i + R_{\mathrm{acc},t}^i + R_{\mathrm{step},t}^i + R_{\mathrm{bdry},t}^i \Big) \end{aligned} \tag{21}$$

The normalization constant $1/50$ keeps the per-step reward within a moderate numerical range, facilitating stable value-function fitting. By construction, the four tiers create complementary gradient signals: the team and progress terms drive convergent approach toward targets, the safety tier shapes repulsive fields around obstacles and teammates, and the auxiliary terms regularize the policy toward smooth, time-efficient trajectories. An empirical analysis of the contribution of each component is provided in Section V-E.

D. Dynamic Assignment-aware MAPPO

Standard MAPPO adopts a Centralized Training with Decentralized Execution (CTDE) paradigm, where a centralized critic guides policy updates but each agent acts on local observations alone. However, when targets move, a fixed or externally determined assignment can quickly become suboptimal, degrading both the critic's value estimates and the actors' navigation performance. To address this, we extend MAPPO with an explicit per-step task allocation mechanism, yielding DA-MAPPO. The key extension is twofold: (i) an online allocator re-solves the assignment problem at every decision step, and (ii) the resulting assignment is injected into each agent's observation, making the policy explicitly conditioned on the current allocation.

Concretely, at each decision time step t, the system observes the current set of UAV positions $\mathcal{P}_t = \{\mathbf{p}_t^i\}_{i=1}^n$ and the set of target positions $\mathcal{Q}_t = \{\mathbf{q}_t^j\}_{j=1}^m$, where $\mathbf{p}_t^i, \mathbf{q}_t^j \in \mathbb{R}^2$. A cost matrix $\mathbf{C}_t \in \mathbb{R}^{n\times m}$ is constructed, whose elements are defined as $C_t^{i,j} = \left\|\mathbf{p}_t^i - \mathbf{q}_t^j\right\|_2^2$, representing the squared Euclidean distance between UAV i and target j. Subsequently, the optimal assignment is obtained by solving the following combinatorial optimization problem:

$$\begin{aligned} \min_{\Pi\in\{0,1\}^{n\times m}} \quad & \sum_{i=1}^{n}\sum_{j=1}^{m} C_t^{i,j}\,\Pi_{i,j} \\ \text{s.t.} \quad & \sum_{j=1}^{m}\Pi_{i,j} = 1\ \forall i,\ \sum_{i=1}^{n}\Pi_{i,j} \le 1\ \forall j \end{aligned} \tag{22}$$

The solution to this optimization problem defines an assignment mapping $\varphi_t : \{1,\cdots,n\} \to \{1,\cdots,m\}$, where $\varphi_t(i)$ denotes the target assigned to UAV i at time step t.

As illustrated in Fig. 2, the raw local observation $\mathbf{o}_t^{\text{raw},i} = [\mathbf{z}_t^i, \mathbf{u}_t^i, \mathbf{q}_t^i]$ is then augmented with the assignment-derived target feature $\mathbf{g}_t^i$ to yield the allocation-augmented observation $\mathbf{o}_t^i = [\mathbf{z}_t^i, \mathbf{u}_t^i, \mathbf{g}_t^i, \mathbf{q}_t^i]$. This augmentation is the critical interface between the allocator and the policy as it makes each agent's action explicitly conditioned on its currently assigned target, enabling the learned policy to seamlessly adapt when assignments change. All agents share a single policy network π_θ, promoting parameter efficiency and behavioral consistency across the swarm.

Following the CTDE paradigm, the shared policy network $\pi_\theta : \mathcal{O} \to \mathcal{A}$ maps each agent's augmented observation to a distribution over actions, while a centralized value network $V_\phi : \mathcal{S} \to \mathbb{R}$ estimates the expected return from the global state. The actor objective combines the PPO clipped surrogate with an entropy bonus to balance exploitation and exploration [52]–[55]:

$$\begin{aligned} \mathcal{L}_{actor}(\theta) = \mathbb{E}_t\Big[& \min\big(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big) \\ & + \sigma\mathcal{H}(\pi_\theta(\cdot|\mathbf{o}_t))\Big] \end{aligned} \tag{23}$$

where the importance sampling ratio is denoted as

$$r_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{o}_t)} \tag{24}$$

with $\mathcal{H}$ representing the policy entropy and σ being the parameter controlling entropy.

The entropy term prevents premature convergence by maintaining stochastic exploration, while the clipping mechanism bounds the policy update magnitude to ensure training stability. The advantage $\hat{A}_t$ is estimated via Generalized Advantage Estimation (GAE), which interpolates between bias and variance through a decay parameter λ:

Algorithm 1 Dynamic Assignment-Aware Multi-Agent PPO

Initialize replay buffer $\mathcal{B}$, policy parameters θ, value function parameters ϕ.
for each training episode **do**
  Reset environment and get initial global state s_0.
  for $t = 0, 1, \ldots, T-1$ **do**
    Observe all agent positions $\{\mathbf{p}_t^i\}_{i=1}^N$ and target positions $\{\mathbf{q}_t^j\}_{j=1}^M$.
    Construct cost matrix $\mathbf{C}_t$.
    Solve the assignment problem to find the optimal mapping $\varphi_t \leftarrow \mathrm{Assign}(\mathbf{C}_t)$.
    Initialize joint action $\mathbf{a}_t \leftarrow \varnothing$.
    for each agent $i = 1, \ldots, N$ **do**
      Obtain assigned target index $j^* = \varphi_t(i)$ and construct extended observation $\mathbf{o}_t^i$.
      Sample action $\mathbf{a}_t^i \sim \pi_\theta(\cdot \mid \mathbf{o}_t^i)$.
      Add $\mathbf{a}_t^i$ to the joint action $\mathbf{a}_t$.
    end for
    Execute joint action $\mathbf{a}_t$, observe reward r_t and next global state s_{t+1}.
    Solve the assignment problem for s_{t+1} and construct the next joint observation $\mathbf{o}_{t+1}$.
    Store transition $(\mathbf{o}_t, \mathbf{a}_t, r_t, \mathbf{o}_{t+1})$ in replay buffer $\mathcal{B}$.
  end for
  Sample a batch of trajectories from $\mathcal{B}$.
  for each time step t in the batch **do**
    Compute advantage estimate $\hat{A}_t$ using critic V_ϕ and global state s_t:
$$\hat{A}_t \leftarrow r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t).$$
  end for
  Update actor parameters $\theta \leftarrow \theta + \alpha\nabla_\theta\mathcal{L}_{\text{actor}}(\theta)$.
  Update critic parameters $\phi \leftarrow \phi - \beta\nabla_\phi\mathcal{L}_{\text{critic}}(\phi)$.
end for

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty}(\gamma\lambda)^l\,\delta_{t+l}^V \tag{25}$$

$$\delta_t^V = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \tag{26}$$

The centralized critic V_ϕ is trained to minimize a clipped mean-squared error against the discounted return, mirroring the conservative update philosophy of the actor:

$$\begin{aligned} \mathcal{L}_{critic}(\phi) = \mathbb{E}_t\Big[\max\Big(& (V_\phi(\mathbf{s}_t) - R_t)^2, \big(\mathrm{clip}(V_\phi(\mathbf{s}_t), \\ & V_{\phi'}(\mathbf{s}_t) - \varepsilon, V_{\phi'}(\mathbf{s}_t) + \varepsilon) - R_t\big)^2\Big)\Big] \end{aligned} \tag{27}$$

where R_t is the discounted return. The clipping on the value function prevents catastrophic overestimation and stabilizes the advantage estimates used by the actor. The complete training procedure, including the interleaving of assignment solving and policy rollouts, is summarized in Algorithm 1.

E. Computational complexity of the online loop

At deployment, each decision step comprises four computations: local information fusion, cost-matrix construction, assignment solution, and policy inference. Summing the fusion workload over all agents yields $O(\sum_{i=1}^{N} k_i)$, which becomes $O(N^2)$ under a fully connected communication graph. Constructing the agent–target cost matrix $\mathbf{C}_t$ requires $O(NM)$ pairwise distance evaluations, whereas solving the rectangular assignment problem in Eq. (22) with a Hungarian-type exact method requires $O(\min(N,M)^2\max(N,M))$. Since the policy network is shared across agents and the per-agent observation dimension is fixed by design, action inference scales linearly with the swarm size, i.e., $O(N)$.

$$\mathcal{T}_{\text{step}} = O\Big(\sum_{i=1}^{N} k_i + NM + \min(N,M)^2\max(N,M) + N\Big). \tag{28}$$

In the balanced setting $M = N$, the per-step complexity reduces to $\mathcal{T}_{\text{step}} = O(N^3+N^2+N) = O(N^3)$, indicating that the assignment module becomes the dominant computational component as the swarm size increases.

V. Experiments and Discussions

A. Experimental Settings

To comprehensively evaluate the performance of the proposed DA-MAPPO algorithm, we establish a unified experimental framework incorporating realistic UAV dynamics, high-fidelity perception, and multi-target mission constraints. All algorithms are implemented within the Gazebo simulation environment, where UAV agents are equipped with simulated LiDAR sensors and operate under local communication constraints. In the Gazebo environment, each UAV is instantiated as a rigid-body model with explicit collision geometry, and all inter-agent and agent–obstacle collisions are handled by the physics engine. The onboard LiDAR measurements are generated through ray casting against the physical bodies, rather than abstract point representations. It should be noted that, although Gazebo enhances physical realism through rigid-body dynamics and collision-aware interactions, environmental disturbances such as wind resistance, aerodynamic drag, and other aerodynamic effects are not explicitly included in the present study. Thus, the reported results are obtained under an ideal windless assumption.

Three widely adopted MARL algorithms are selected for comparison. To broaden the comparison beyond MARL variants, we additionally include a RL-based navigation method tailored for dynamic obstacle avoidance (NavRL) and one state-of-the-art optimization-based local planner (EGO-Planner v2) as representative non-MARL baselines:

1) Independent PPO(IPPO) [56]: Fully decentralized PPO in which each agent learns independently without shared critic information.
2) Multi-Agent PPO(MAPPO) [52]: A state-of-the-art CTDE-based algorithm with a centralized critic providing stronger global value estimation.
3) Recurrent Multi-Agent PPO (RMAPPO) [52]: A recurrent extension of MAPPO designed to better capture temporal dependencies in partially observable environments.
4) NavRL [57]: A PPO-based RL navigation framework designed for safe flight in cluttered dynamic environments. NavRL uses tailored state/action representations for collision avoidance and additionally applies a lightweight safety shield inspired by velocity-obstacle concepts to project potentially unsafe velocity commands back into a safe region during deployment.
5) EGO-Planner v2 [58]: A representative optimization-based local trajectory planner for aerial swarms that performs fast spatial-temporal trajectory optimization under multi-objective costs and safety constraints, producing dynamically feasible trajectories within milliseconds.

TABLE I
Training platform specifications

Component	Specs
CPU	AMD Ryzen 9 7950X
GPU	NVIDIA GeForce RTX 4090
RAM	64 GB DDR5
OS	Ubuntu 18.04 LTS
Python	3.6.1
PyTorch	1.5.1 + CUDA 10.1

TABLE II
Algorithm hyperparameters

Hyperparameter	Value
MLP layers	3
Hidden dimension	256
Learning rate	1×10^{-5}
PPO epochs	10
Number of mini-batches	1
Clipping parameter	0.2
Entropy coefficient	0.1
Discount factor	0.99
Episode length	600
Total environment steps	3×10^{6}

All algorithms are trained using a unified hardware platform and software environment, with specific configurations detailed in Table I. To ensure a fair comparison, the trained algorithms employ identical network architectures and hyperparameter settings, as specified in Table II.

To facilitate stable convergence and scalable performance, a progressive curriculum learning strategy is used. The number of static obstacles increases gradually from fewer than 10 in the early training phase to 35–40 in the final stage, as detailed in Table III.

To quantitatively evaluate the performance of different swarm formation control algorithms, five metrics are adopted. Let N_{total} denote the total number of evaluation episodes, N_{success} the number of successful episodes, $N_{\text{collision}}$ the number of episodes terminated by inter-UAV collisions, and N_{timeout} the number of episodes that exceed the maximum allowed steps.

- Mission Success Rate (R_{success}) is the percentage of episodes in which all UAVs successfully reach their assigned targets, i.e., $R_{\text{success}} = N_{\text{success}}/N_{\text{total}}$. It is the

TABLE III
TRAINING ENVIRONMENT CONFIGURATION

Steps	Obstacles
$0 \sim 4.0 \times 10^5$	0–10
$4.0 \times 10^5 \sim 8.0 \times 10^5$	10–20
$8.0 \times 10^5 \sim 1.2 \times 10^6$	20–25
$1.2 \times 10^6 \sim 1.6 \times 10^6$	25–30
$1.6 \times 10^6 \sim 2.0 \times 10^6$	30–35
$2.0 \times 10^6 \sim 3.0 \times 10^6$	35–40

TABLE IV
EXPERIMENTAL RESULTS OF UAV SWARM NAVIGATION IN STATIC MULTI-TARGET SCENARIOS

Scenarios	Algorithms	Metrics				
		R_{success}	$R_{\text{collision}}$	R_{timeout}	T_{ave}	L_{ave}
ENV-1: 30 obstacles	IPPO	94	6	0	**36.18**	17.96
	RMAPPO	98	2	0	36.83	18.20
	MAPPO	98	2	0	36.66	17.94
	NavRL	83	17	0	48.81	18.70
	Ego-Planner v2	87	13	0	41.57	**16.61**
	DA-MAPPO	**99**	**1**	**0**	37.14	17.87
ENV-2: 40 obstacles	IPPO	91	9	0	**37.95**	18.69
	RMAPPO	93	7	0	39.87	19.53
	MAPPO	92	8	0	41.40	19.99
	NavRL	83	17	0	48.81	18.70
	Ego-Planner v2	77	22	1	43.59	**17.24**
	DA-MAPPO	**95**	**5**	**0**	38.03	18.25
ENV-3: 50 obstacles	IPPO	87	13	0	**39.18**	19.20
	RMAPPO	87	13	0	43.03	20.96
	MAPPO	89	11	0	44.19	21.07
	NavRL	41	54	0	51.00	18.95
	Ego-Planner v2	57	41	2	46.08	**17.52**
	DA-MAPPO	**92**	**8**	**0**	40.34	19.40

primary indicator of overall swarm coordination capability.

- Collision Rate ($R_{\text{collision}}$) is the percentage of episodes terminated due to inter-UAV or UAV–obstacle collisions. It reflects the safety performance of the swarm under different controllers.
- Timeout Rate (R_{timeout}) is the percentage of episodes in which the swarm fails to complete the mission within the maximum number of steps.
- Average Trajectory Length (L_{ave}) is the mean total path length traveled by all UAVs across successful episodes. A shorter L_{ave} indicates more direct and efficient flight paths.
- Average Time Steps (T_{ave}) is the mean number of decision steps required to complete the mission across successful episodes. Together with L_{ave}, it characterizes the operational efficiency of the swarm.

B. Static Multi-Target Navigation Experiments

In static multi-target navigation experiments, the target locations remain fixed throughout each episode, while UAVs are required to simultaneously reach distinct target points in cluttered environments. This setting evaluates the fundamental swarm coordination and collision avoidance capability. Three static environments with increasing obstacle density are considered, including ENV-1 (30 obstacles), ENV-2 (40 obstacles), and ENV-3 (50 obstacles). For each environment, 100 independent trials are conducted to obtain statistically meaningful performance metrics.

TABLE V
EXPERIMENTAL RESULTS OF UAV SWARM NAVIGATION IN DYNAMIC MULTI-TARGET SCENARIOS

Scenarios	Algorithms	Metrics				
		R_{success}	$R_{\text{collision}}$	R_{timeout}	T_{ave}	L_{ave}
ENV-1: 30 obstacles	IPPO	78	22	0	44.84	21.49
	RMAPPO	85	15	0	43.49	21.49
	MAPPO	83	17	0	44.41	21.49
	NavRL	63	34	3	54.02	20.46
	Ego-Planner v2	53	45	2	47.33	18.68
	DA-MAPPO	**99**	**1**	**0**	**37.16**	**18.01**
ENV-2: 40 obstacles	IPPO	57	43	0	46.15	22.38
	RMAPPO	69	31	0	46.01	22.71
	MAPPO	70	30	0	50.14	23.96
	NavRL	53	43	4	53.97	20.45
	Ego-Planner v2	51	48	1	48.42	19.08
	DA-MAPPO	**95**	**5**	**0**	**38.32**	**18.50**
ENV-3: 50 obstacles	IPPO	53	47	0	48.75	22.75
	RMAPPO	67	33	0	50.41	24.71
	MAPPO	64	36	0	54.52	26.02
	NavRL	32	65	3	55.37	20.91
	Ego-Planner v2	43	57	0	48.48	19.73
	DA-MAPPO	**90**	**10**	**0**	**40.88**	**19.35**

Table IV reports the quantitative results. DA-MAPPO consistently achieves the best overall performance, yielding the highest success rates and the lowest collision rates across all environments. In ENV-1, DA-MAPPO attains a 99% success rate. As obstacle density increases, its performance remains robust (92% success in ENV-3), whereas MAPPO and RMAPPO degrade to 89% and 87%, respectively. NavRL and EGO-Planner v2 exhibit pronounced performance drops in dense scenes (41% and 57% success in ENV-3), indicating that single-agent RL and optimization-based local planning are insufficient to resolve strong multi-agent couplings without explicit coordination.

In terms of efficiency, DA-MAPPO maintains competitive average steps and trajectory lengths across obstacle densities. Although EGO-Planner v2 occasionally yields shorter paths due to aggressive local optimization, this advantage is offset by substantially higher collision and failure rates in cluttered settings. In contrast, DA-MAPPO achieves a more favorable safety–efficiency trade-off, producing smooth trajectories while preserving high task completion rates. The relatively stable T_{ave} and L_{ave} further suggest that its robustness does not depend on excessive detours or overly conservative behaviors as the environment becomes more complex.

Figure 4 qualitatively corroborates these findings. IPPO exhibits oscillatory motions and near-collision behaviors, reflecting limited coordination among independently trained agents. MAPPO and RMAPPO improve coordination but still show trajectory overlaps and local congestion in dense regions. NavRL often generates locally feasible yet globally suboptimal behaviors, leading to bottlenecks or deadlocks when free space is contested. EGO-Planner v2 produces smooth single-agent trajectories but lacks swarm-level awareness, resulting in frequent inter-agent conflicts. By contrast, DA-MAPPO yields well-separated and coordinated trajectories, enabling proactive adjustments to avoid both obstacles and neighboring agents, consistent with the improved success and collision statistics in Table IV.

Overall, the static experiment results demonstrate that explicit coordination under the CTDE framework is critical

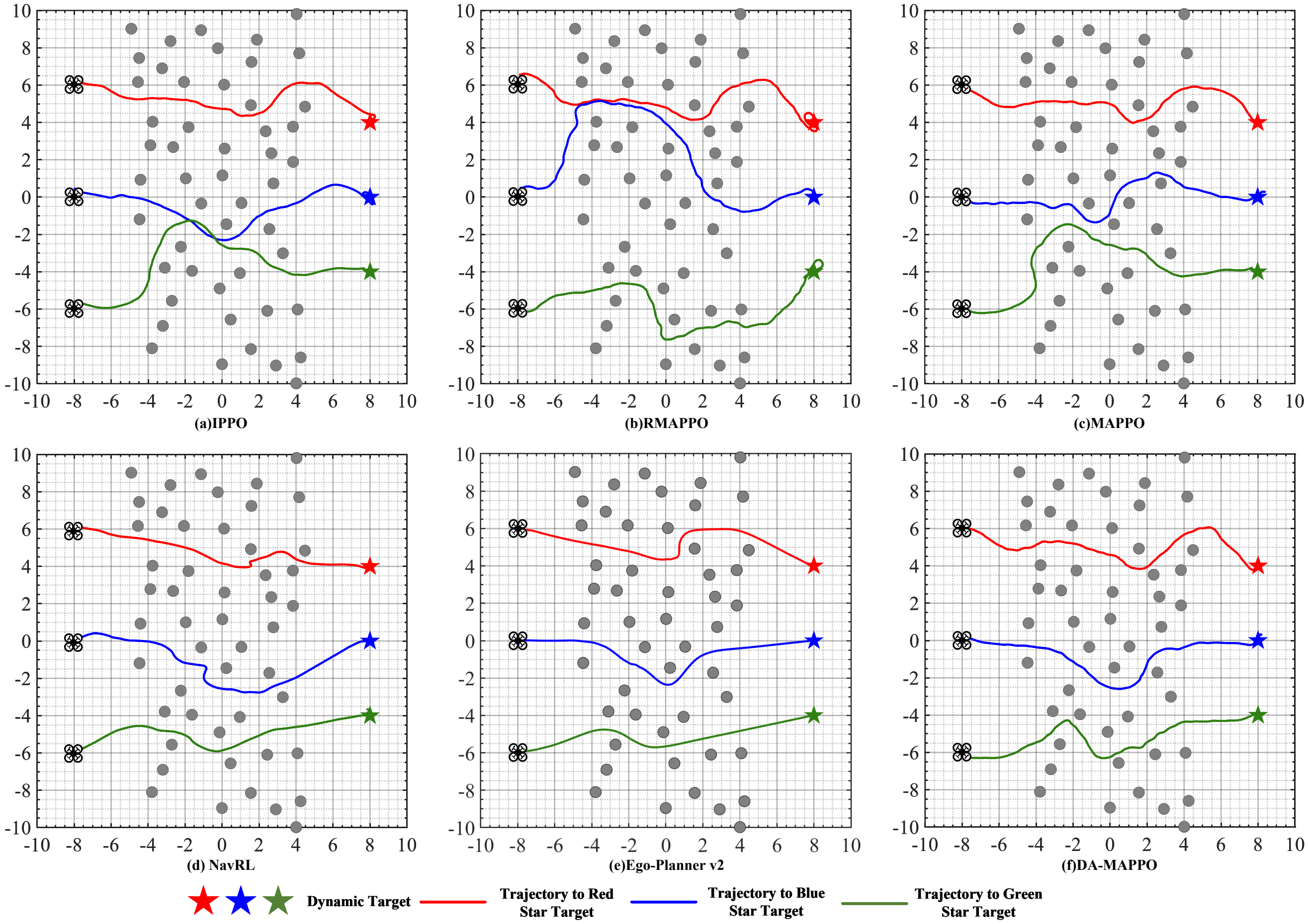

Fig. 4. Swarm Navigation Schematic Diagram of Different Algorithms in Static Multi-Target Scenarios.

for scalable multi-target swarm navigation. While centralized critics improve performance over fully decentralized methods, they remain limited when target allocation and collision avoidance must be jointly addressed. By integrating dynamic assignment-aware observations into a unified multi-agent policy, DA-MAPPO effectively bridges the gap between coordination efficiency and safety, achieving robust performance even in densely cluttered environments.

C. Dynamic Multi-Target Navigation Experiments

In the dynamic multi-target navigation experiments, the positions of the target points dynamically change during the course of the mission. Specifically, the target points are swapped with other target points, simulating real-world scenarios where target assignments are not fixed. This dynamic reassignment introduces additional complexity and challenges in swarm coordination, requiring continuous adaptation from the UAVs. The experimental environments remain the same as in the static navigation tasks.

Table V reveals that DA-MAPPO's advantage over all baselines is substantially amplified when targets become dynamic. DA-MAPPO achieves 99%, 95%, and 90% success in ENV-1 through ENV-3, respectively, with collision rates of at most 10%. In contrast, the MARL baselines (IPPO, MAPPO, RMAPPO) suffer 15–47% collision rates, whereas NavRL and EGO-Planner v2 method drop to 32–63% success in ENV-3. Notably, DA-MAPPO's success rate in the dynamic setting is nearly identical to its static counterpart (e.g., 99% vs. 99% in ENV-1, 90% vs. 92% in ENV-3), whereas all baselines exhibit substantially larger static-to-dynamic degradations. This resilience directly validates the benefit of per-step online allocation.

DA-MAPPO simultaneously achieves the best efficiency, attaining the lowest T_{ave} and L_{ave} across all environments (see Table V). Baselines without dynamic reallocation incur longer paths due to the pursuit of outdated targets, while EGO-Planner v2's locally optimal trajectories are offset by frequent collision-induced failures.

The qualitative trajectories in Fig. 5 reinforce these findings. The MARL baselines produce redundant, erratic paths because rigid assignments force UAVs to pursue suboptimal goals under evolving conditions. NavRL and EGO-Planner v2 generate smoother local paths but lack swarm-level spatial awareness, resulting in frequent inter-agent conflicts. DA-MAPPO produces the most coordinated behavior: real-time reallocation enables proactive congestion resolution, yielding well-separated trajectories even in densely cluttered, dynamic environments.

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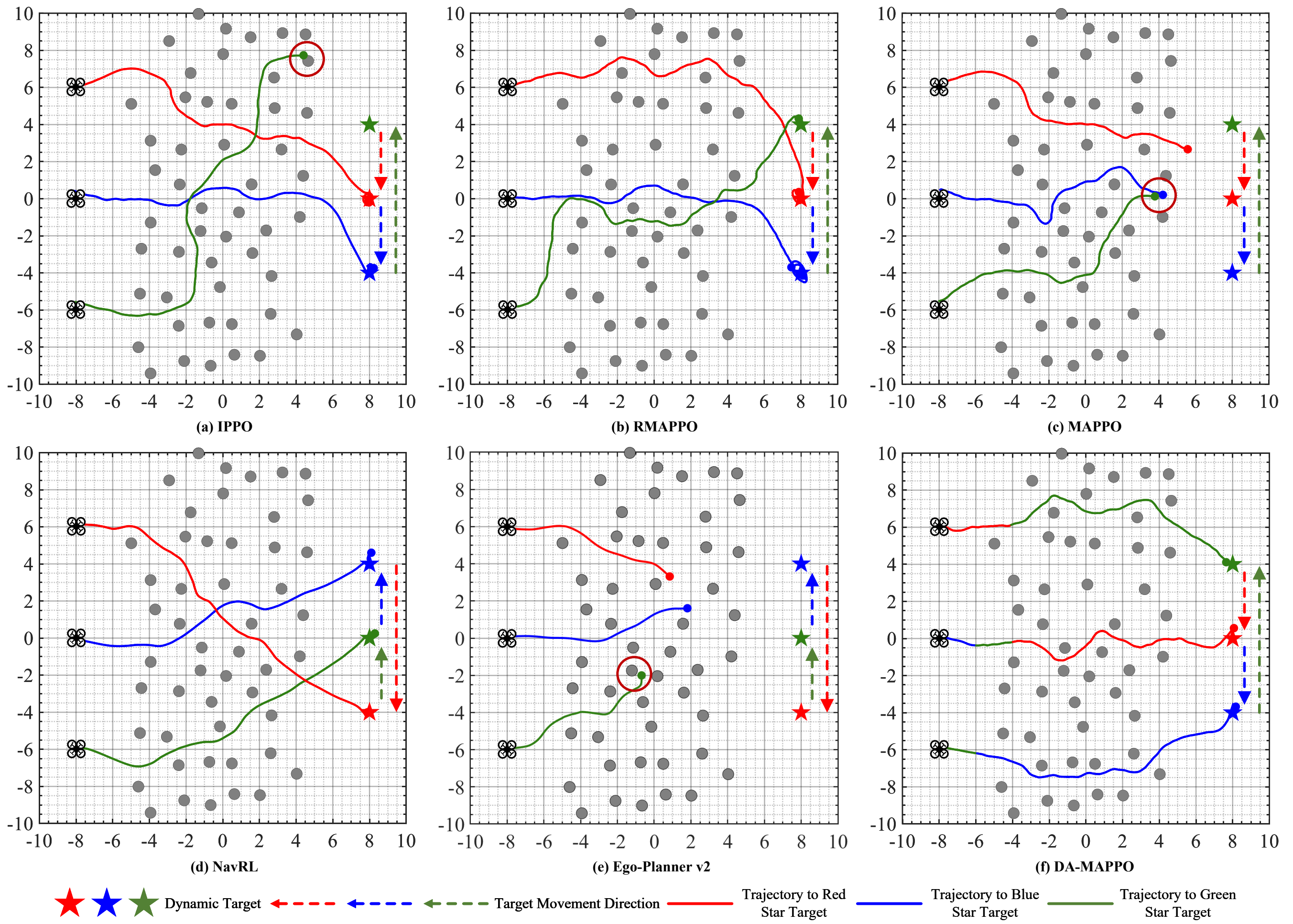

Fig. 5. Swarm navigation schematic diagram in dynamic scenarios. The colors of the UAV trajectories (solid lines) are matched to the color of their currently assigned targets (stars) at each decision step. Note that in panel (f), a color change in a single UAV's path indicates a real-time target reassignment executed by the DA-MAPPO framework to optimize team-level efficiency.

D. Ablation Studies

To validate the effectiveness of key design components in the proposed DA-MAPPO algorithm, we systematically evaluate three ablated variants:

1) DA-MAPPO (50 Steps): Reduces the dynamic target allocation frequency to once every 50 steps.
2) w/o team reward: Removes the team distance reward term during training.
3) w/o augmented state: Removes the augmented state while the dynamic allocator continues to pair agents with targets during training.

Experiments are conducted in dynamic multi-target scenarios, with results summarized in Table VI.

Reducing the frequency of target reallocation in DA-MAPPO (50 Steps) leads to a noticeable decrease in performance across all environments. In ENV-1 (30 obstacles), this variant achieves 96% success compared to 99% with the full DA-MAPPO, while the collision rate increases, highlighting the importance of frequent target reallocation for maintaining optimal coordination and minimizing collisions. This suggests that dynamic target reassignment at every step is critical for maintaining high success rates in dynamic environments.

Removing the team reward causes a 4–6% success rate decline and higher collision rates, indicating that the shared potential signal plays a meaningful role in maintaining spatial separation and coordinated progress. This confirms that maintaining proper spatial separation through team rewards is vital for efficient and collision-free multi-UAV coordination.

To further isolate the specific contribution of the assignment-augmented observation state, a variant named w/o augmented state was evaluated. As shown in Table VI, the success rate for this variant is 0%, with agents failing to reach any targets. This result demonstrates that the proposed DA-MAPPO framework is not merely driven by heuristic target guidance. Instead, the policy network has learned a critical dependency on the assignment-augmented state g_t^i to perceive its mission objective. Without this explicit information, the agents are unable to establish a functional navigation policy, confirming that the integration of dynamic assignment results into the MARL observation space is a prerequisite for successful swarm coordination in dynamic IoT environments.

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TABLE VI
ABLATION STUDY RESULTS IN DYNAMIC MULTI-TARGET SCENARIOS

Environments	Algorithms	Metrics				
		R_{success}	$R_{\text{collision}}$	R_{timeout}	T_{ave}	L_{ave}
ENV-1: 30 obstacles	w/o team reward	95	5	0	37.15	18.11
	w/o augmented state	0	100	0	–	–
	DA-MAPPO (50Steps)	96	4	0	37.72	18.47
	DA-MAPPO	99	1	0	37.16	18.01
ENV-2: 40 obstacles	w/o team reward	93	7	0	38.96	18.75
	w/o augmented state	0	100	0	–	–
	DA-MAPPO (50Steps)	90	10	0	39.72	19.34
	DA-MAPPO	94	6	0	38.49	18.72
ENV-3: 50 obstacles	w/o team reward	84	16	0	41.10	19.68
	w/o augmented state	0	100	0	–	–
	DA-MAPPO (50Steps)	85	15	0	41.19	19.99
	DA-MAPPO	90	10	0	40.88	19.35

TABLE VII
REWARD COMPONENT STATISTICS BY TERMINATION (SUCCESS EPISODES)

Component	Mean	Abs Percent of total (%)	Role/Note
Team	-688.4	10.3	Global distance
Distance progress	3784.1	56.5	Main positive signal
Hover	24.9	0.5	After arrival
Drift	-8.9	0.1	After arrival
Arrived bonus	675.0	10.1	Time-ranked arrival
Crash	-446.8	6.7	Obstacle collision
Inter-uav crash	0.0	0.0	Rare in eval
Lidar	-0.6	0.0	Proximity
Step penalty	-577.3	8.6	Time
Acceleration	-483.8	7.2	Smoothness
Boundary	0.0	0.0	No violation in eval
All scaled	45.8	–	(all components)/50

TABLE VIII
REWARD COMPONENT STATISTICS BY TERMINATION (COLLISION EPISODES)

Component	Mean	Abs Percent of total (%)	Role/Note
Team	-573.9	14.1	Global distance
Distance progress	2304.6	56.8	Main positive signal
Hover	0.0	0.0	After arrival
Drift	0.0	0.0	After arrival
Arrived bonus	0.0	0.0	Time-ranked arrival
Crash	-561.7	13.8	Obstacle collision
Inter-uav crash	0.0	0.0	Rare in eval
Lidar	-2.3	0.1	Proximity
Step penalty	-344.0	8.5	Time
Acceleration	-274.4	6.8	Smoothness
Boundary	0.0	0.0	No violation in eval
All scaled	11.0	–	(all components)/50

E. Reward Contribution Analysis

To better understand how each reward term shapes the learned behavior, we perform a post-hoc reward contribution analysis on the trained policy. During evaluation, we record, for each episode, the cumulative reward of every component aggregated over all three drones and all time steps, together with the scaled total return. The episodes are then grouped by termination type (SUCCESS vs. COLLISION). For each group, we compute the mean value of every reward component and its absolute contribution share. The results are summarized in Tables VII and VIII.

Several clear insights can be drawn from these results. First, in both SUCCESS and COLLISION episodes, distance progress is the dominant positive term, contributing 56.5% and 56.8% of the total absolute reward magnitude, respectively. This confirms that forward progress toward the goal is indeed the principal driving signal of the policy, consistent with the intended reward design.

Second, SUCCESS episodes are distinguished by the presence of a substantial arrived bonus, while this term is completely absent in COLLISION episodes. At the same time, SUCCESS episodes still incur non-negligible penalties from team distance, step penalty, and acceleration. This indicates that successful behavior is not achieved by exploiting a single reward term, but rather by balancing task completion with formation maintenance, time efficiency, and motion smoothness.

Third, COLLISION episodes exhibit markedly stronger negative influence from safety-related terms. In particular, the crash penalty and team penalty account for a larger fraction of the total reward magnitude in COLLISION episodes than in SUCCESS episodes. Although the agents may still accumulate considerable distance-progress reward before failure, these penalties, together with the missing arrival bonus, substantially reduce the final scaled return.

Overall, the scaled total return is much higher for SUCCESS episodes than for COLLISION episodes. This clear separation shows that the reward function provides well-differentiated feedback between desirable and undesirable behaviors. Therefore, the analysis supports the design rationale in Section IV: the reward is dominated by progress toward the goal, while auxiliary penalties and terminal terms effectively bias the learned policy toward safe, coordinated, and efficient navigation.

F. Robustness Analysis

Real-world IoT-edge deployments inevitably involve sensor noise, communication impairments, and varying environmental dynamics. We evaluate DA-MAPPO's robustness by these experiments:

TABLE IX
EXPERIMENTAL RESULTS UNDER VARYING INTENSITIES OF NOISE INTERFERENCE

Metrics	$\sigma_v = 0.20$ $\sigma_\omega = 0.05$	$\sigma_v = 0.30$ $\sigma_\omega = 0.10$	$\sigma_v = 0.50$ $\sigma_\omega = 0.15$	Baseline
R_{success}	94	94	90	95
$R_{\text{collision}}$	6	6	10	5
R_{timeout}	0	0	0	0
T_{ave}	39.14	39.29	40.28	38.32
L_{ave}	18.83	19.44	20.11	18.50

TABLE X
EXPERIMENTAL RESULTS UNDER VARYING INTENSITIES OF COMMUNICATION INTERFERENCE

Metrics	$\eta = 0.10$ $\tau = 1$ step	$\eta = 0.30$ $\tau = 3$ steps	$\eta = 0.50$ $\tau = 6$ steps	Baseline
R_{success}	94	94	94	95
$R_{\text{collision}}$	6	6	6	5
R_{timeout}	0	0	0	0
T_{ave}	38.40	38.57	38.53	38.32
L_{ave}	18.56	18.65	18.63	18.50

- Sensor noise. Additive Gaussian noise of increasing intensity is injected into velocity and angular velocity observations at test time. As shown in Table IX, success rate degrades gracefully from 95% (baseline) to 90% at the highest noise level, with only a modest increase in collision rate and path length. The policy's tolerance to perceptual noise can be attributed to the Kalman-filtered acceleration estimates and the redundancy provided by multi-modal observations.
- Communication interference. Stochastic packet loss (η up to 0.50) and latency (τ up to 6 steps) are introduced into the inter-UAV channel. Table X shows that all performance metrics remain virtually unchanged, confirming that the decentralized execution paradigm does not critically depend on perfect, instantaneous information exchange.
- Computational feasibility. Inference time is profiled on an NVIDIA Jetson Orin Nano edge platform (Fig. 6). The per-step latency scales sub-linearly with agent count, confirming that the framework supports high-frequency execution on resource-constrained IoT-edge hardware.
- Target dynamics sensitivity. To stress-test the allocation–policy coupling, the target moving speed is increased from the nominal 0.5 m/s up to 3 m/s (6 times the UAV's maximum speed) while keeping the allocation frequency fixed at 5 Hz. As shown in Table XI, R_{success} remains constant at 95% across all speeds, with only marginal increases in T_{ave} and L_{ave}. This indicates that the per-step allocation mechanism provides sufficient reactivity without requiring a higher update rate, even under aggressive target dynamics.

Taken together, these results demonstrate that DA-MAPPO degrades gracefully under sensor noise, communication impairments, and fast-moving targets, while remaining computationally feasible for real-time onboard deployment—key prerequisites for practical IoT-edge swarm operations.

TABLE XI
THE EXPERIMENTAL RESULTS OF VARYING MOVING SPEEDS AT TARGET POINTS

Metrics	0.5 m/s	1 m/s	3 m/s
R_{success}	95	95	95
$R_{\text{collision}}$	5	5	5
R_{timeout}	0	0	0
T_{ave}	38.32	38.62	38.90
L_{ave}	18.50	18.57	18.69

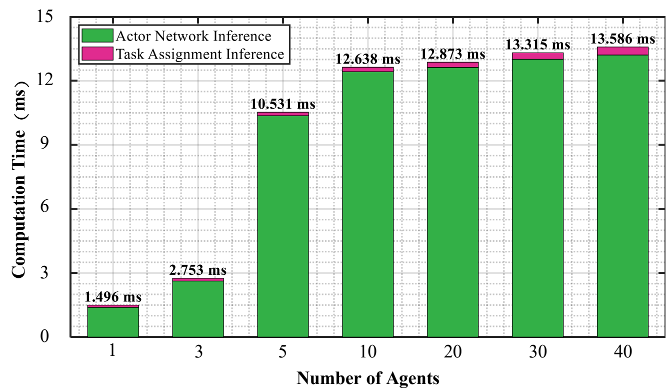

Fig. 6. Total Algorithm Execution Time per Step.

VI. CONCLUSION

This work addresses the challenge of tightly coupling task allocation and cooperative navigation for UAV swarms operating at the IoT edge under dynamic targets, partial observability, and constrained communication. To this end, we proposed DA-MAPPO, a dynamic-assignment-aware multi-agent reinforcement learning framework that integrates online target allocation with decentralized policy learning. A hierarchical cooperative reward was further designed to jointly promote navigation efficiency, swarm coordination, and operational safety.

Comprehensive experiments in high-fidelity Gazebo environments demonstrate that DA-MAPPO consistently outperforms representative baselines in both static and dynamic multi-target scenarios. In dynamic environments with varying obstacle densities, DA-MAPPO achieves high success rates of 90%–99%, while maintaining low collision rates and competitive trajectory efficiency. Notably, it shows only marginal performance degradation when targets change from static to dynamic, demonstrating strong adaptability and robustness in complex IoT-edge settings.

Future work will focus on real-world UAV deployment for sim-to-real validation and on communication-efficient coordination under stricter bandwidth and latency constraints.

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